

Relativistic Quantum Field Theory I(PH.60053)

Problem Set 1

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Note on units: I generally disdain the use of natural units in undergraduate studies. For students having gone through a symbolic mess of a calculation, something as simple as dimensional analysis gives a good sanity check. And while this course is explicitly designed for graduate students, I would like to stand by this conviction and perform all calculations with explicit engineering dimensions and constants. I believe it is reasonably simple for the grader to convert it into natural units. I appreciate your understanding.

During class, the professor has instructed us that the engineering dimensions of ϕ must be determined such that $\mathcal{L}$ has dimensions $[EL^{-3}]$. This means $[\phi] = [E^{1/2}L^{-1/2}]$, and the Klein-Gordon equation is given by

$$\left(\square + \left(\frac{mc}{\hbar}\right)^2\right)\phi = \left(-\frac{1}{c^2}\partial_t^2 + \vec{\nabla}^2 + \left(\frac{mc}{\hbar}\right)^2\right)\phi = 0.$$

Suppose further that the mode expansion is given as:

$$\phi(t, \vec{x}) = \int \frac{d^3\vec{p}}{(2\pi\hbar)^{3/2}} \mathcal{N}(E_{\vec{p}}) \left(a_{\vec{p}} e^{\frac{i\vec{p}\cdot\vec{x}}{\hbar}} + a_{\vec{p}}^\dagger e^{-\frac{i\vec{p}\cdot\vec{x}}{\hbar}} \right),$$

where $x^\mu = (ct, \vec{x})$, $p^\mu = (E_{\vec{p}}/c, \vec{p})$, and $E_{\vec{p}} = \sqrt{(mc^2)^2 + (\vec{p}c)^2}$. Insisting the simple commutation relation:

$$[a_{\vec{p}}, a_{\vec{p}'}] = \delta^{(3)}(\vec{p} - \vec{p}')$$

implies that the normalization factor $\mathcal{N}(E_{\vec{p}})$ has dimensions $[E^{1/2}L]$, or

$$d\Pi_{\vec{p}} := \frac{d^3\vec{p}}{(2\pi\hbar)^{3/2}} \mathcal{N}(E_{\vec{p}}) = \frac{d^3\vec{p} \hbar c}{(2\pi\hbar)^{3/2} \sqrt{2E_{\vec{p}}}}.$$

Problem 3.1

Let us substitute the mode expansion into the definition of the Gaussian-smeared operator:

$$A(a) = \frac{1}{(a\sqrt{\pi})^3} \int d\Pi_{\vec{p}} \int d^3\vec{x} \left(a_{\vec{p}} e^{\frac{i\vec{p}\cdot\vec{x}}{\hbar}} + a_{\vec{p}}^\dagger e^{-\frac{i\vec{p}\cdot\vec{x}}{\hbar}} \right) e^{-\vec{x}^2/a^2}.$$

Here, because the time coordinate is zero, $p \cdot x = \vec{p} \cdot \vec{x}$. Thus, we can integrate the exponentials over the spatial coordinates first.

$$\begin{aligned}
\int d^3\vec{x} \exp\left(-\frac{\vec{x}^2}{a^2} + \frac{i\vec{p} \cdot \vec{x}}{\hbar}\right) &= \int_0^\infty dr r^2 e^{-r^2/a^2} \int_{-1}^1 d(\cos\theta) \int_0^{2\pi} d\phi e^{\frac{i|\vec{p}|r}{\hbar} \cos\theta} \\
&= \int_0^\infty dr r^2 e^{-r^2/a^2} \cdot 4\pi \frac{\sin \frac{|\vec{p}|r}{\hbar}}{|\vec{p}|r/\hbar} \\
&= \frac{4\pi\hbar}{|\vec{p}|} \int_0^\infty dr r \sin \frac{|\vec{p}|r}{\hbar} e^{-r^2/a^2} \\
&= \frac{4\pi\hbar}{|\vec{p}|} \left(-\hbar \frac{\partial}{\partial(|\vec{p}|)}\right) \int_0^\infty dr \cos \frac{|\vec{p}|r}{\hbar} e^{-r^2/a^2} \\
&= \frac{4\pi\hbar^2}{|\vec{p}|} \left(-\frac{\partial}{\partial(|\vec{p}|)}\right) \frac{1}{2} \operatorname{Re} \int_{-\infty}^\infty dr \exp\left(-\frac{r^2}{a^2} + \frac{i|\vec{p}|r}{\hbar}\right) \\
&= \frac{2\pi\hbar^2}{|\vec{p}|} \left(-\frac{\partial}{\partial(|\vec{p}|)}\right) \operatorname{Re} \int_{-\infty}^\infty dr \exp\left(-\frac{1}{a^2} \left(r - \frac{i|\vec{p}|a^2}{\hbar}\right)^2 - \frac{|\vec{p}|^2 a^2}{4\hbar^2}\right) \\
&= \frac{2\pi\hbar^2}{|\vec{p}|} \left(-\frac{\partial}{\partial(|\vec{p}|)}\right) \sqrt{\pi} a \cdot \exp\left(-\frac{|\vec{p}|^2 a^2}{4\hbar^2}\right) \\
&= (a\sqrt{\pi})^3 \exp\left(-\frac{|\vec{p}|^2 a^2}{4\hbar^2}\right) \\
&\Rightarrow A(a) = \int d\Pi_{\vec{p}} \left(a_{\vec{p}} + a_{\vec{p}}^\dagger\right) \exp\left(-\frac{|\vec{p}|^2 a^2}{4\hbar^2}\right)
\end{aligned}$$

Since $a_{\vec{p}}|0\rangle = \langle 0|a_{\vec{p}}^\dagger = 0$, this shows that $\langle 0|A(a)|0\rangle = 0$.

$$\begin{aligned}
\Rightarrow \operatorname{var} A(a) &= \langle 0|A(a)^2|0\rangle \\
&= \int d\Pi_{\vec{p}} \int d\Pi_{\vec{p}'} \langle 0| \left(a_{\vec{p}}a_{\vec{p}'} + a_{\vec{p}}a_{\vec{p}'}^\dagger + a_{\vec{p}}^\dagger a_{\vec{p}'} + a_{\vec{p}}^\dagger a_{\vec{p}'}^\dagger\right) |0\rangle \exp\left(-\frac{(|\vec{p}|^2 + |\vec{p}'|^2)a^2}{4\hbar^2}\right) \\
&= \int d\Pi_{\vec{p}} \int d\Pi_{\vec{p}'} \langle 0|a_{\vec{p}}a_{\vec{p}'}^\dagger|0\rangle \exp\left(-\frac{(|\vec{p}|^2 + |\vec{p}'|^2)a^2}{4\hbar^2}\right) \\
&= \int d\Pi_{\vec{p}} \int d\Pi_{\vec{p}'} \langle 0| \left(a_{\vec{p}'}^\dagger a_{\vec{p}} + \delta^{(3)}(\vec{p} - \vec{p}')\right) |0\rangle \exp\left(-\frac{(|\vec{p}|^2 + |\vec{p}'|^2)a^2}{4\hbar^2}\right) \\
&= \int \frac{d^3\vec{p}}{(2\pi\hbar)^3} \frac{\hbar^2 c^2}{2E_{\vec{p}}} \exp\left(-\frac{|\vec{p}|^2 a^2}{2\hbar^2}\right) \\
&= \int \frac{d^3\vec{\kappa}}{(2\pi\hbar)^3} \frac{\hbar^2 c^2 (\hbar/a)^3}{2\hbar c/a \sqrt{\vec{\kappa}^2 + (mca/\hbar)^2}} e^{-\vec{\kappa}^2/2} \quad (\vec{\kappa} := \vec{p}a/\hbar) \\
&= \frac{\hbar c}{16\pi^3 a^2} \int d^3\vec{\kappa} \frac{e^{-\vec{\kappa}^2/2}}{\sqrt{\vec{\kappa}^2 + (mca/\hbar)^2}} \\
&= \frac{\hbar c}{4\pi^2 a^2} \int_0^\infty d\kappa \frac{\kappa^2 e^{-\kappa^2/2}}{\sqrt{\kappa^2 + (mca/\hbar)^2}}
\end{aligned}$$

Let us define the dimensionless parameter $\mu := mca/\hbar$ and find the asymptotic behavior:

- $\mu \gg 1$:

$$\begin{aligned}
\text{var } A(a) &= \frac{\hbar c}{4\pi^2 a^2} \int d\kappa \frac{\kappa^2}{\mu} e^{-\kappa^2/2} \left(1 + \left(\frac{\kappa}{\mu} \right)^2 \right)^{-1/2} \\
&= \frac{\hbar c}{4\pi^2 a^2 \mu} \int d\kappa e^{-\kappa^2/2} \left(\kappa^2 - \frac{\kappa^4}{2\mu^2} + \dots \right) \\
&\sim \frac{\hbar c}{\sqrt{32\pi^3} a^2 \mu} (1 + O(\mu^{-2})) \\
&= \left(\frac{\hbar^2}{\sqrt{32\pi^3} m} \right) a^{-3} \cdot \left(1 + O\left(\left(\frac{mca}{\hbar} \right)^{-2} \right) \right)
\end{aligned}$$

Smearing over regions much larger than the correlation length of the field, the average fluctuation inside the region scales like the volume of the region. This can be understood in terms of field correlations decaying to zero at long distances, rendering the fluctuations essentially independent between separated regions.

- $\mu \ll 1$:

$$\begin{aligned}
\text{var } A(a) &= \frac{\hbar c}{4\pi^2 a^2} \int d\kappa \kappa e^{-\kappa^2/2} \left(1 + \left(\frac{\mu}{\kappa} \right)^2 \right)^{-1/2} \\
&= \frac{\hbar c}{4\pi^2 a^2} \int d\kappa e^{-\kappa^2/2} \left(\kappa - \frac{\mu^2}{2\kappa} + \dots \right) \\
&\sim \frac{\hbar c}{4\pi^2 a^2} (1 + O(\mu^2)) \\
&= \left(\frac{\hbar c}{4\pi^2} \right) a^{-2} \cdot \left(1 + O\left(\left(\frac{mca}{\hbar} \right)^2 \right) \right)
\end{aligned}$$

In regions much smaller than the correlation length, any fluctuation of the field inside the region must have a short wavelength, i.e., a large momentum of order $\hbar/a \gg mc$. Thus, the particles are approximately massless, and the field fluctuation follows an inverse-area law a^{-2} .

Problem 3.2

(a) For a complex scalar field $\phi : \mathbb{R}^{1,3} \rightarrow \mathbb{C}$, we assert that the Lagrangian density $\mathcal{L}[\phi]$ must satisfy the following conditions:

- **Lorentz invariance:** Upon any Poincaré transform (a, Λ) , the Lagrangian density must satisfy $\mathcal{L}[\phi] \xrightarrow{(a, \Lambda)} \mathcal{L}'[\phi'] = \mathcal{L}[\phi]$.
- **Reality:** $\mathcal{L}^\dagger = \mathcal{L}$ (abusing the dagger to denote complex conjugation as well).
- **Global $U(1)$ symmetry:** The Lagrangian density must be a function only of the ray of ϕ , as the vectors in the ray correspond to the same physical state. That is, it must be invariant under $\phi \mapsto e^{i\alpha}\phi$.
- **Locality:** The Lagrangian density must be a function only of powers of ϕ , derivatives of ϕ , or their complex conjugate.

With the lowest nontrivial order 2 for each ϕ and $\partial_\mu\phi$, one naturally obtains the free field theory:

$$\mathcal{L} = -\partial_\mu\phi^\dagger\partial^\mu\phi - \left(\frac{mc}{\hbar}\right)^2\phi^\dagger\phi.$$

(The *free field* qualifier will be verified below by the mode expansions.)

For the purposes of calculus of variation to obtain the equations of motion, we must treat $\phi(t, \vec{x})$ and $\phi^\dagger(t, \vec{x})$ as independent degrees of freedom. Note that we still have the Klein-Gordon equation:

$$0 = \partial^\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi^\dagger)} \right) - \frac{\partial \mathcal{L}}{\partial \phi^\dagger} = - \left(\square + \left(\frac{mc}{\hbar} \right)^2 \right) \phi.$$

(b) The Lagrangian density can be expanded as:

$$\mathcal{L} = \left(\frac{1}{c^2} \dot{\phi}^\dagger(t, \vec{x}) \dot{\phi}(t, \vec{x}) - \vec{\nabla} \phi^\dagger(t, \vec{x}) \cdot \vec{\nabla} \phi(t, \vec{x}) - \left(\frac{mc}{\hbar} \right)^2 \phi^\dagger(t, \vec{x}) \phi(t, \vec{x}) \right)$$

The canonical momenta are found as:

$$\pi(t, \vec{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}} = \frac{\dot{\phi}^\dagger(t, \vec{x})}{c^2}, \quad \pi^\dagger(t, \vec{x}) = \frac{\partial \mathcal{L}}{\partial \dot{\phi}^\dagger} = \frac{\dot{\phi}(t, \vec{x})}{c^2}.$$

With these, we can write down the canonical commutation relations:

$$\begin{aligned} [\phi(t, \vec{x}), \phi(t, \vec{y})] &= [\phi(t, \vec{x}), \phi^\dagger(t, \vec{y})] = [\phi^\dagger(t, \vec{x}), \phi^\dagger(t, \vec{y})] = 0, \\ [\pi(t, \vec{x}), \pi(t, \vec{y})] &= [\pi(t, \vec{x}), \pi^\dagger(t, \vec{y})] = [\pi^\dagger(t, \vec{x}), \pi^\dagger(t, \vec{y})] = 0, \\ [\phi(t, \vec{x}), \pi^\dagger(t, \vec{y})] &= [\phi^\dagger(t, \vec{x}), \pi(t, \vec{y})] = 0, \\ [\phi(t, \vec{x}), \pi(t, \vec{y})] &= [\phi^\dagger(t, \vec{x}), \pi^\dagger(t, \vec{y})] = i\hbar\delta^{(3)}(\vec{x} - \vec{y}). \end{aligned}$$

(Note: The last two relations are consistent, as $[\phi^\dagger, \pi^\dagger] = -[\phi, \pi]^\dagger$.) The same mode expansion schema as the free real scalar field theory lets us write:

$$\phi(t, \vec{x}) = \int d\Pi_{\vec{p}} \left(a_{\vec{p}} e^{\frac{ip \cdot x}{\hbar}} + b_{\vec{p}}^\dagger e^{-\frac{ip \cdot x}{\hbar}} \right), \quad \phi^\dagger(t, \vec{x}) = \int d\Pi_{\vec{p}} \left(b_{\vec{p}} e^{\frac{ip \cdot x}{\hbar}} + a_{\vec{p}}^\dagger e^{-\frac{ip \cdot x}{\hbar}} \right),$$

with $E_{\vec{p}}$ defined exactly the same as before. Because we no longer insist that ϕ be Hermitian ($\phi^\dagger = \phi$), we cannot conclude that $a_{\vec{p}} = b_{\vec{p}}$ and must treat them together. The same inverse Fourier (mode?) transform allows us to write:

$$\begin{aligned} a_{\vec{p}} &= \int \frac{d^3\vec{x} \hbar c}{(2\pi\hbar)^{3/2} \sqrt{2E_{\vec{p}}}} \left(E_{\vec{p}} \phi(t, \vec{x}) + ic^2 \pi^\dagger(t, \vec{x}) \right) e^{-\frac{ip \cdot x}{\hbar}}, \\ b_{\vec{p}}^\dagger &= \int \frac{d^3\vec{x} \hbar c}{(2\pi\hbar)^{3/2} \sqrt{2E_{\vec{p}}}} \left(E_{\vec{p}} \phi(t, \vec{x}) - ic^2 \pi^\dagger(t, \vec{x}) \right) e^{\frac{ip \cdot x}{\hbar}}, \\ b_{\vec{p}} &= \int \frac{d^3\vec{x} \hbar c}{(2\pi\hbar)^{3/2} \sqrt{2E_{\vec{p}}}} \left(E_{\vec{p}} \phi^\dagger(t, \vec{x}) + ic^2 \pi(t, \vec{x}) \right) e^{-\frac{ip \cdot x}{\hbar}}, \\ a_{\vec{p}}^\dagger &= \int \frac{d^3\vec{x} \hbar c}{(2\pi\hbar)^{3/2} \sqrt{2E_{\vec{p}}}} \left(E_{\vec{p}} \phi^\dagger(t, \vec{x}) - ic^2 \pi(t, \vec{x}) \right) e^{\frac{ip \cdot x}{\hbar}}. \end{aligned}$$

Even more laborious calculations yield the commutation relations of the mode operators:

$$[a_{\vec{p}}, a_{\vec{p}'}] = [a_{\vec{p}}, a_{\vec{p}'}^\dagger] = [a_{\vec{p}}^\dagger, a_{\vec{p}'}^\dagger] = [b_{\vec{p}}, b_{\vec{p}'}] = [b_{\vec{p}}, b_{\vec{p}'}^\dagger] = [b_{\vec{p}}^\dagger, b_{\vec{p}'}^\dagger] = 0,$$

$$[a_{\vec{p}}, b_{\vec{p}'}] = [a_{\vec{p}}, b_{\vec{p}'}^\dagger] = [a_{\vec{p}}^\dagger, b_{\vec{p}'}] = [a_{\vec{p}}^\dagger, b_{\vec{p}'}^\dagger] = 0,$$

$$[a_{\vec{p}}, a_{\vec{p}'}^\dagger] = [b_{\vec{p}}, b_{\vec{p}'}^\dagger] = \delta^{(3)}(\vec{p} - \vec{p}').$$

Thus, this algebra corresponds to *two* independent harmonic oscillators per continuous label $\vec{p}$! In part (d), we will be able to show that the corresponding particles of a and b will carry opposite-sign charges, justifying their relationship as *antiparticles*.